Candidate Ranking Analysis Report

Project Context:
Madhuri, Program Manager at Tech Enterprises, needed to urgently fill a critical resource gap due to a team member resigning. The HR team, led by Ritesh, sourced internal candidates and shared applications with Madhuri. The objective was to rank candidates based on multiple criteria to identify the most suitable candidate efficiently.

Dataset Overview

The dataset contains the following candidate information:

Column	Description
Employee Name	Name of the candidate
Year of Experience	Total work experience in years
Appraisal 1, 2, 3	Historical appraisal ratings (0–1 scale)
Skills	List of skills possessed
Key Projects	List of key projects handled
Duration in Current Role	Time spent in the current role (years)
Bench Duration	Time spent on bench (years)
When the Candidate Will Be Available	Expected joining date
Derived Columns	Days until availability, Appraisal history average, Count of Skills, Count of Key Projects

Preprocessing Steps:

1. Converted "When the candidate will be available" into days remaining from reference date **1st July 2024**.
2. Computed **Appraisal History** as the average of appraisals above cutoff 0.7.
3. Counted the number of skills and key projects for each candidate.

Ranking Methodology

Step 1: Normalization

To compare heterogeneous criteria on a common scale, all numeric criteria were normalized using **Min-Max Normalization**:

$$\text{Normalized Value} = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

- **Positive criteria (higher is better):** Years of experience, Appraisal history, Skills count, Key Projects count, Duration in current role, Bench duration.
- **Negative criteria (lower is better):** Days until availability.

Step 2: Composite Score Calculation

Two methods were used to calculate a **composite score** for each candidate:

1. **Method 1 – Equal Weights:** All criteria weighted equally (each criterion 1/7).
2. **Method 2 – Madhuri's Weighted Preferences:** Criteria weighted as per the given table:

Criteria	**Weight**
Years of Experience	20%
Appraisal History	10%
Skills	20%
Key Projects	10%
Duration in Current Role	10%
Bench Duration	10%
Days Until Available	20%

Composite Score formula:

$$\text{Composite Score} = \sum(\text{Normalized Value} \times \text{Weight})$$

Step 3: Ranking

- Candidates were ranked **descendingly** based on composite scores.
- **Rank = 1** corresponds to the highest composite score.

Results

Candidate	Rank (Method 1 – Equal Weights)	Rank (Method 2 – Weighted)
Abhijit	1	2
Lavanya	8	9
Siva	3	3
Akanksha	4	4
Sazid	6	6
Anuj	7	7
Praveen	5	5
Esha	2	1
Nanda	9	10
Shalini	10	8

Observations:

1. **Akanksha** retains the **same rank** (4) in both methods.
2. **Siva** also retains the same rank (3) across both methods.
3. Overall, **2 candidates** retain the same rank in both methods.
4. Method 2 rankings shift slightly for candidates whose strengths align more with Madhuri's weighted preferences.

Key Insights

1. **Top Candidates:**
    - **Method 1:** Abhijit scored highest based on equal weighting.
    - **Method 2:** Esha scored highest due to weighted criteria favoring experience, skills, and availability.
2. **Impact of Weighting:**
    - Weighted ranking gives priority to strategic importance (e.g., availability and critical skills).
    - Equal weighting may undervalue certain critical attributes.

3. **Normalization Necessity:**
    - Required to bring heterogeneous data (years, counts, days) onto a **comparable scale**.
    - Prevents any single metric from dominating the composite score due to different units.
4. **Derived Columns Added:**
    - Days until availability
    - Appraisal History
    - Count Skills
    - Count Key Projects

Conclusion

- **Decision-making using weighted ranking** ensures alignment with project priorities.
- Normalization and composite scoring are essential for **fair multi-criteria evaluation**.
- From the analysis:
    - **Esha** is the top candidate if Madhuri's weighted preferences are followed.
    - **Akanksha** and **Siva** retain consistent rankings across both methods.
- This approach is **scalable** and can be used for future candidate evaluation in Tech Enterprises.